A Quantitative Framework for Regime Detection Using Slope Analysis and Zigzag Extremes

PCTT Methodology Upgrade v2

Robust derivatives, adaptive structure, probabilistic regimes, and change-point detection for live trading systems

Version: 2026-01-23

Abstract

Markets switch between regimes (trend, range, transition, volatility expansion) faster than most lagging indicators can track. This document upgrades a slope + zigzag approach into an implementable, mathematically robust regime detector suitable for live PCTT deployments across stocks, futures, options (underlying-driven), crypto, and FX. Key upgrades: robust derivatives on log-price, volatility normalization, adaptive non-repainting zigzag pivots, structure-aware features, probabilistic regimes (score-based or HMM), online change-point detection, multi-timescale fusion, and integration into PCTT scoring and risk. This is not a promise of profitability; it is a testable framework for more stable and explainable regime detection.

1. Robust Slope Estimation on Log-Price

1.1 Local polynomial regression (endpoint derivative)

Fit a polynomial of order k on the last W samples of log-price $x_t = \log(P_t)$: $x_{t-i} \approx a0 + a1\, i + a2\, i^2 + \dots + ak\, i^k$. Then slope at the endpoint is $m_t = a1$ and curvature is $c_t = 2a2$. With fixed coefficients this is Savitzky–Golay filtering, which is fast and streamable.

1.2 Robustification against spikes

Outliers can dominate least squares. Implementable choices: robust loss (Huber/Tukey), winsorized returns, or downweight points with $|r|>q\cdot\sigma$. This reduces false acceleration signals during gaps/news.

1.3 Optional Kalman smoothing

Use a local linear trend state-space model where slope is latent. Kalman filtering yields smoothed slope plus uncertainty, enabling confidence-aware regime probabilities.

2. Volatility-normalized Slope, Curvature, and Confidence

2.1 EWMA volatility

Compute EWMA volatility on returns: $\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1-\lambda) r_t^2$, with $\lambda \approx 0.94$ intraday and 0.97–0.99 swing.

2.2 Normalized slope and curvature

Define $m■_t = m_t / \sigma_t$ and $c■_t = c_t / \sigma_t$. These are dimensionless signal-to-noise ratios comparable across instruments and timeframes. Large $|m■|$ indicates stronger trend; $\mathrm{sign}(c■)$ indicates acceleration vs deceleration.

2.3 Regression t-stat for trend vs chop

Compute $t_m = \beta / SE(\beta)$ from a window regression. Use $|t_m|$ as confidence gating and as an HMM emission feature.

3. Adaptive Zigzag Extremes (Non-repainting)

3.1 Volatility-scaled pivot threshold

Use $\theta_t = \kappa \cdot ATR_t$ (price space) or $\theta_t = \kappa \cdot \sigma_t \cdot \mathrm{sqrt}(\Delta t)$ (log space). κ is the structure sensitivity.

3.2 Confirmed pivot semantics

Maintain a pivot candidate and confirm only after retracement by $\geq \theta_t$. Confirmed pivots never repaint; the current candidate may update until confirmed.

3.3 Structure features

From pivots compute swing returns, durations, angles, pivot density, and HH/HL vs LH/LL dominance. These features anchor regimes to confirmed structure.

4. Feature Fusion and Regime Inference

Form an observation vector o_t combining momentum + structure. Example:

```
o_t = [m■_t, c■_t, t_m, ER_t, S_hhhl,t, swingAngle_t, pivotDensity_t]
```

Compute ER_t (efficiency ratio) to separate trend from chop, and S_hhhl,t as a signed structure-quality score from HH/HL dominance.

4.1 Score-based regime classifier

Compute scores per regime and convert to probabilities via softmax. This is the simplest deployable model and provides explainability.

4.2 Hidden Markov Model (recommended upgrade)

Use a small HMM with regimes {UpTrend, DownTrend, Range, Transition, VolExpansion}. High diagonal transition probabilities enforce persistence. Forward filtering yields stable probabilities and fewer whipsaws than threshold-only logic.

5. Change-point Detection (Transition prior)

Add CUSUM or Page–Hinkley on normalized returns/slope to detect distribution shifts early. When triggered, boost Transition probability and reduce risk until stabilization.

6. Multi-timescale Fusion

Compute the same feature set on multiple timeframes (e.g., 5m/1h/4h). Fuse with weights favoring higher timeframes or require agreement. This prevents micro-noise from flipping the regime against higher-timeframe structure.

7. Integration into PCTT

Add slope component S_s and regime component S_r to the PCTT composite score Q_t. Gate entries by regime probability and slope sign. Scale risk by confidence and tighten stops when curvature flips against position direction.

8. Implementation Examples

8.1 Streaming zigzag (pseudocode)

```
state:
  candidate = (idx, price, type)
  threshold = kappa * ATR

on_new_bar(idx, high, low, close):
  update threshold using ATR/vol

  if candidate.type == HIGH:
     if high > candidate.price: candidate = (idx, high, HIGH)
     if close < candidate.price - threshold:
        confirm pivot HIGH
        candidate = (idx, low, LOW)
  else:
     if low < candidate.price: candidate = (idx, low, LOW)
     if close > candidate.price + threshold:
        confirm pivot LOW
        candidate = (idx, high, HIGH)
```

8.2 Python slope/curvature

```python
import numpy as np

def poly_slope_curvature(log_prices: np.ndarray, k: int = 2):
    W = len(log_prices)
    i = np.arange(W)
    X = np.vander(i, N=k+1, increasing=True)
    a, *_ = np.linalg.lstsq(X, log_prices, rcond=None)
    slope = float(a[1])
    curvature = float(2*a[2]) if k >= 2 else 0.0
    return slope, curvature
```

8.3 TypeScript EWMA vol

```typescript
export function ewmaVol(returns: number[], lambda = 0.94): number {
  let s = 0;
  for (let i = 0; i < returns.length; i++) {
    s = lambda * s + (1 - lambda) * returns[i] * returns[i];
  }
  return Math.sqrt(s);
}
```

8.4 Pine normalized slope proxy

```
//@version=5
indicator("PCTT Slope Regime (v2)", overlay=false)
len = input.int(20, "Window")
x = math.log(close)
lin = ta.linreg(x, len, 0)
m = lin - lin[1]
vol = ta.stdev(ta.change(x), len)
m_norm = m / math.max(vol, 1e-6)
plot(m_norm, "Normalized Slope")
hline(0)
```

9. Live Visual Explainability (the “wow” layer)

Overlay confirmed zigzag pivots and swing legs; shade background by regime; render a probability ribbon and slope/curvature panels; annotate change-points. Generate a short agent narrative in a side panel explaining the drivers and risk adjustments.